Audit-Gated Gradient Projection from Information-Curvature Index Subspaces for Continual Learning

Y.Y.N. Li

Abstract

Catastrophic forgetting arises when gradient updates for a new task overwrite parameter directions critical to a previously learned task. We argue that the information field tensor Γ^{info}—a curvature object derived from the entropy functional of the model's predictive distribution [12]—provides a geometry-informed signal for continual learning: directions in the approximate null space of Γ^{info} are information-neutral and potentially safe to update.

We instantiate this view through an *audit-gated gradient projection family*. Rather than claiming exact full-parameter null-space recovery, we use *Gamma cross-batch reproducibility audits*: each parameter's Γ^{info} sub-block is estimated on two disjoint batch halves, and only parameters whose near-null eigenspaces align across both halves (Criterion A > 0.5, passing in ≥ 2 independent subsets) enter gradient projection. A matched random-direction control (same support indices and subspace rank) isolates whether the *audit-identified direction*—not merely the projection operation—is the source of any forgetting benefit.

In a cross-domain continual learning experiment (GPT-2, WikiText-2 $\rightarrow$ Biomedical Medical QA), the audit gates 41–42 of 42 candidate parameters across 5 random seeds, demonstrating robust null-space structure throughout GPT-2 layers h.6–h.11. Audit-gated null projection (`gamma_along`) significantly reduces forgetting versus unconstrained fine-tuning ($+331 \pm 30$ vs. $+414 \pm 45$, $\Delta = -83$, $p<0.05$, 5 seeds), while preserving Task B perplexity (9.55 vs. 9.54 for free). The direction signal is directionally present: `gamma_along` $<$ `gamma_random` ($\Delta = -38$), supporting the geometric claim that audit-identified null directions—not merely projection—reduce forgetting.

1 Introduction

Continual learning (CL) faces the fundamental challenge of *catastrophic forgetting*: fine-tuning on a new task (Task B) degrades performance on a previously learned task (Task A) [3, 9]. The dominant approach, Elastic Weight Consolidation (EWC) [4], penalizes updates to parameters deemed important via a diagonal Fisher approximation. While effective in practice, EWC provides no geometric account of *which directions in parameter space* are safe to update—it treats parameter importance as a scalar rather than a directional, layer-aware property.

The geometry question. We take a different starting point: does the parameter space of a trained neural network have intrinsic geometric structure that *predicts* forgetting risk before any Task B training occurs? If so, that structure could guide CL strategy—identifying which layers to protect, which directions to project out, and how to scale learning rates—all from a single pre-training audit.

The information field equation. Li [12] establish that the information field tensor Γ^{info}—derived from the entropy functional of the model's predictive distribution—captures this structure. They show empirically on GPT-2 and ViT-Tiny that $\|\Gamma^{\text{info}}\|$ decays monotonically with depth (ratio $\approx$ 20:1 from block 0 to block 11), that $\|\Gamma^{\text{info}}\|$ predicts per-layer forgetting (Spearman $\rho = 0.822 \pm 0.054$), and that geometry-guided per-layer learning rates outperform uniform schedules. This makes Γ^{info} a *predictive geometry*: a pre-training signal that encodes

layer stability. Yet a layer-level signal only tells us *which layers* are at risk; it does not identify *which directions within those layers* can be updated safely. The null space of Γ^{info}—directions along which information curvature vanishes—is precisely that directional complement.

This paper. We extend the predictive geometry of Γ^{info} to the null-space perspective. Our contributions are distinct from Li [12] along three axes: (i) *From high-curvature to zero-curvature geometry*: Li characterise layers where $\|\Gamma^{\text{info}}\|$ is large (high forgetting risk); we exploit the complementary regime where $\Gamma^{\text{info}}v = 0$ (safe update directions). (ii) *From layer-level signal to direction-level control*: Li predict which layers are dangerous; we identify which *directions within those layers* can be updated safely. (iii) *From predictive signal to audit-time actionable mechanism*: Li provide a forgetting predictor; we provide the Criterion A audit and Stage-2 projection that act on that prediction before Task B training begins. Our contributions are:

1. A Gamma cross-batch reproducibility audit (Criterion A) that extends Li [12]'s layer-level stability signal to direction-level null-space verification before CL, with quantitative thresholds (§4).

2. Empirical verification on GPT-2 (WikiText-2 → Biomedical QA, 5 seeds) showing audit-gated null projection significantly reduces forgetting ($p<0.05$) without sacrificing Task B performance (§5).

3. Evidence that null *direction quality* matters: `gamma_along` outperforms matched random projection, supporting the geometric interpretation (§5).

2 Background

2.1 Continual Learning and Catastrophic Forgetting

Let $\theta \in \mathbb{R}^d$ denote model parameters. After fine-tuning on Task A to obtain θ_A, we wish to fine-tune on Task B while minimizing Task A forgetting:

$$\Delta\mathcal{L}_A = \mathcal{L}_A(\theta_{AB}) - \mathcal{L}_A(\theta_A),$$

where θ_{AB} are parameters after Task B training.

2.2 Elastic Weight Consolidation

EWC adds a quadratic penalty [4]:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_B(\theta) + \frac{\lambda}{2}\sum_i F_i(\theta_i - \theta_{A,i})^2,$$

where F_i is the i-th diagonal Fisher entry. EWC is scalar per parameter and does not encode directional geometry.

3 Information Field Tensor and Its Null Space

3.1 The Information Field Equation

Let $p_\theta(y \mid x)$ denote the model's predictive distribution. The entropy functional is:

$$H(\theta) = -\mathbb{E}_{x\sim\mathcal{D}}\left[\sum_y p_\theta(y \mid x)\log p_\theta(y \mid x)\right].$$

We define the *information field tensor* as the second-order covariant derivative of H with respect to θ:

$$\Gamma_{ij}^{\text{info}}(\theta) = \frac{\partial^2 H}{\partial\theta_i \partial\theta_j} - \sum_k \Gamma_{ij}^k \frac{\partial H}{\partial\theta_k}, \tag{1}$$

where Γ_{ij}^k are Christoffel symbols of the Fisher–Rao metric. In the flat (Euclidean) approximation standard in deep learning, $\Gamma_{ij}^k \approx 0$ and Γ^{info} reduces to the Hessian of H.

While Eq. 1 gives the general geometric definition, in practice we work in the Euclidean approximation ($\Gamma_{ij}^k \approx 0$). Li [12] derive an exact four-term decomposition of the K=1 information-time metric $G_{ij} = \nabla_i \nabla_j (dt_{\text{info}}^2)$:

$$G = T_1 + T_2 + T_3 + T_4,$$

where T_1 is the Fisher metric term, T_2 is the entropy curvature term $\nabla^2 H$, and T_3, T_4 are cross and higher-order residuals. They show empirically (corr = 1.000, 12/12 sub-blocks on ViT-Tiny) that when T_2 dominates (87–114% of $\|G\|$)—the entropy-dominated regime—the field equation reduces to:

$$G_{ij} \approx -2\, dt_{\text{info}}^2\, \Gamma_{ij}^{\text{info}}, \tag{2}$$

with $\Gamma_{ij}^{\text{info}} = (\nabla_i \nabla_j H)/H - 2(\nabla_i H)(\nabla_j H)/H^2$ the information curvature tensor (a normalized form of Eq. 1 under the Euclidean approximation, scaled by H^{-1}). The approximation achieves corr = 0.988 ± 0.006 on ViT-Tiny (3 seeds) and corr > 0.975 on GPT-2 interior blocks (layers 2–9). Crucially, Li [12] report accuracy *degrades* at boundary blocks where T_1 is non-negligible—a finding that directly motivates caution at boundary blocks; our audit covers layers `h.6`–`h.11` and treats any failures at outer layers as expected (see §4).

Important: the reduction from the full information-time metric G to the curvature tensor Γ^{info} is an empirical regime approximation, not an algebraic identity. Throughout this paper we therefore treat Γ^{info} as an audit-time geometric proxy whose validity must be verified experimentally; Criterion A is precisely designed to perform that verification before CL training begins.

3.2 $\|\Gamma^{\text{info}}\|$ as a Layer Stability Signal

Beyond the null space, the *magnitude* $\|\Gamma_l^{\text{info}}\|$ of the information curvature at layer l carries predictive information about that layer's behavior under fine-tuning. Li [12] show on GPT-2 that $\|\Gamma_l^{\text{info}}\|$ decays monotonically with depth (ratio $\approx$ 20:1 from early to late blocks) and that it correlates strongly with per-layer forgetting (Spearman $\rho = 0.822 \pm 0.054$) and anti-correlates with learning capacity ($\rho = -0.807 \pm 0.046$).

This motivates the following principle:

> *High-curvature layers ($\|\Gamma_l^{\text{info}}\|$ large) are geometrically unstable: they forget more and learn less efficiently. Low-curvature layers are geometrically stable and are the natural target for null-space projection.*

If this principle holds, parameters in low-curvature regions should exhibit higher cross-batch reproducibility (A) of their Γ^{info} near-null subspaces. Our audit provides exactly such a test.

Corroboration from our own audit. Consistent with the predictive geometry picture, layer-norm and attention parameters—which reside in lower-curvature regions of the transformer block—exhibit near-perfect cross-batch reproducibility ($A \approx 1.0$ for `ln.weight` and `attn.c_attn.bias`; Table 1). The only audit failures occur in the largest MLP weight matrices (`mlp.c_fc.weight`, $d = 2.36$M, pass rate 90%): higher dimensionality makes null-space estimation inherently noisier at fixed subspace size $k \leq 50$, so cross-batch reproducibility is harder to achieve. This pattern is directionally aligned with the depth-stability findings of Li [12], though six parameter types are too few for a statistical claim.

The predictive geometry picture further motivates curvature-aware extensions to null-space projection, formalized in §7.

3.3 Null Space Definition

Definition 1 (Γ-null space)**.** *The null space of* Γ^{info} *at* θ *is:*

$$\mathcal{N}(\Gamma^{\text{info}})(\theta) = \{v \in \mathbb{R}^d : \Gamma^{\text{info}}(\theta)\, v = 0\}.$$

Directions in $\mathcal{N}(\Gamma^{\text{info}})$ *are* information-neutral*: the* ***second-order*** *entropy curvature vanishes along* v *(see Remark 1); the first-order term is governed by the stationarity of* θ*.*

Remark 1 (Second-order entropy neutrality of null directions)**.** *If* $v \in \mathcal{N}(\Gamma^{\text{info}})(\theta_A)$ *and the Euclidean approximation holds so that* Γ^{info} *reduces to the entropy Hessian* $\nabla^2 H$*, then for small* ε*:*

$$H(\theta_A + \varepsilon v) = H(\theta_A) + \varepsilon \nabla H(\theta_A)^\top v + O(\varepsilon^3).$$

That is, the ***second-order*** *entropy curvature vanishes along* v *(by definition of the null space under the Euclidean approximation). The first-order term vanishes when* θ_A *is approximately stationary, which holds after Task A fine-tuning.*

Remark 2. *Remark 1 is a* second-order *result: null directions suppress entropy curvature but not the first-order drift. In practice, after fine-tuning on Task A,* θ_A *is approximately stationary (*$\|\nabla H(\theta_A)\| \approx 0$*), so the first-order residual is small; we treat this as an empirical condition rather than a theoretical guarantee.*

This implies that gradient updates confined to $\mathcal{N}(\Gamma^{\text{info}})$ do not alter the *curvature* of the entropy landscape to second order, providing a theoretical basis for reduced forgetting under the stationarity assumption.

3.4 Subspace Estimation

Computing the exact null space of Γ^{info} is infeasible for large parameters (d up to 2.36M in our experiments). In practice, Γ^{info} is therefore estimated over a random k-dimensional subspace via:

$$\hat{\Gamma^{\text{info}}} = Q^\top H_\theta^{\text{ent}} Q,$$

where $Q \in \mathbb{R}^{d \times k}$ is a random orthonormal projection matrix ($k = 50$ in our experiments) and H_θ^{ent} is the empirical *entropy* Hessian computed over N probe batches. Null directions are eigenvectors of $\hat{\Gamma^{\text{info}}}$ with eigenvalue below threshold τ (default $\tau = 0.05$). Note that $H_\theta^{\text{ent}} \neq H_\theta^{\text{loss}}$ in general; Criterion A mitigates this by requiring cross-batch reproducibility of the near-null subspace rather than agreement with the loss Hessian directly.

4 The Alignment Audit Framework

Given a Task A checkpoint θ_A, our audit evaluates null-space quality along Criterion A before CL training begins (Algorithm 1). As established in §3, the field equation is an empirical regime approximation that may degrade at boundary blocks [12]; we audit all target layers (`h.6`–`h.11`) and treat lower A scores at outer layers as consistent with this prediction.

Scope: index-subspace audit. Criterion A is evaluated over randomly sampled coordinate subsets of size k (default $k = 50$), operating on $k \times k$ sub-blocks of Γ^{info}. This is a *local index-subspace audit*—it verifies that certain sampled coordinate subspaces admit reliable near-null surrogate directions, but does not constitute a global proof of null-space validity for the full parameter tensor. The multi-subset gate ($\geq M_{\min} = 2$) guards against single-subset coincidences.

Algorithm 1 Audit-Gated Gamma Cross-Batch Null Extraction (Stage 1)

Require: Task A checkpoint θ_A, probe dataset $\mathcal{D}_{\text{probe}}$, threshold A^*, null tolerance τ

```
Fine-tune on Task A (if not already done)
for each target parameter θ_i do
    for each random index subset s = 1, ..., N_subsets do
        Sample k indices; split N_batches into two halves
        Compute k×k sub-blocks Γ̂info^(1,s), Γ̂info^(2,s) on each half
        Evaluate Criterion A (Eq. 3): cross-batch near-null alignment
    end for
end for
Mark θ_i as audit-positive if A > A* in ≥ M_min subsets
Store best-subset null basis (highest-A passing subset) for audit-positive θ_i
return Audit-positive parameters → Stage 2
```

4.1 Criterion A: Gamma Cross-Batch Reproducibility

Let near-null(M) denote eigenvectors of M with $|\lambda| < \tau\|M\|_F$. Γ^{info} is estimated on two disjoint batch halves from the same Task A data: $\hat{\Gamma^{\text{info}}}^{(1)}$ from the first $\lfloor N/2 \rfloor$ batches, $\hat{\Gamma^{\text{info}}}^{(2)}$ from the remainder.

$$A = \text{mean}\ \cos^2\!\big(\text{near-null}(\hat{\Gamma^{\text{info}}}^{(1)}),\ \text{near-null}(\hat{\Gamma^{\text{info}}}^{(2)})\big). \tag{3}$$

A measures cross-batch reproducibility of the near-null eigenspace of Γ^{info}. High A (> 0.5) implies that the near-null subspace is a stable geometric property of the parameter space under Task A, not a data artifact. Unlike earlier formulations that required a loss-Hessian sub-block (H_sub), an NCCL structural proxy [10] and a gradient-conditioned cost ratio (RC), Criterion A requires only Task A data and is free of loss-Hessian or gradient-conditioning dependencies. Passes when $A > A^* = 0.5$ in $\geq M_{\min} = 2$ independent subsets.

4.2 Audit Gate

The multi-subset requirement guards against single-subset coincidences. Only audit-positive parameters enter Stage 2. The stored projection basis is the near-null vectors from the subset with the highest A (most reproducible).

5 Experiments

5.1 Setup

Model. GPT-2 (124M parameters, 12 transformer layers).

Tasks (cross-domain). Task A: WikiText-2 (general text; 2 epochs fine-tuning, lr $= 5 \times 10^{-4}$). Task B: Biomedical Medical QA (`medical_meadow_mediqa`; 800 train / 100 val chunks). Forgetting metric: $\Delta\text{ppl}_A = \text{ppl}_A^{\text{after}} - \text{ppl}_A^{\text{before}}$ on held-out WikiText-2 test chunks. The domain gap (general $\rightarrow$ biomedical) stresses forgetting more severely than same-domain WikiText-2/103 transfer and thus provides a more discriminative benchmark for CL methods.

Stage 1 (Audit). Audit target: layers `h.6`–`h.11` (42 candidate parameters). Criterion A: Gamma cross-batch reproducibility—Γ^{info} sub-blocks are estimated on two disjoint batch halves ($N_{\text{batches}} = 5$, split 3|2); near-null alignment between both halves is measured by principal-angle overlap. Configuration: $k \leq 50$ subspace dimensions ($k = 50$ for $d \leq 1024$, $k = 16$ for $d \leq 4096$, $k = 12$ for $d > 4096$), $N_{\text{subsets}} = 4$, threshold $A^* = 0.5$, $\tau = 0.15$, $M_{\min} = 2$. Two-pass hybrid finite-difference correction handles Conv1D parameters with near-zero autograd HVP rows.

Stage 2 (CL). Fine-tune on Task B: 3 epochs, lr $= 5 \times 10^{-4}$, batch size 8. Seeds: $\{42, 123, 456, 789, 1234\}$; all results reported as mean $\pm$ std over 5 seeds.[1] Significance: paired two-tailed t-test vs. free across seeds.

Baselines. We compare against representative regularization- and masking-based baselines:

- **free**: unconstrained fine-tuning on Task B.
- **EWC** [4]: Fisher-weighted L^2 penalty.
- **oEWC** [6]: online EWC (diverges at lr $= 5 \times 10^{-4}$; reported as failed).
- **packnet_simplified**: one-shot magnitude-based gradient mask (bottom 50% weights frozen); simplified PackNet [5].
- **gamma_random**: matched random-direction control—same support indices and subspace rank as `gamma_along`, random orthonormal basis. Isolates direction quality from projection geometry.
- **gamma_along** ($\alpha = 1.0$): gradient projected fully onto audit-gated Γ^{info} null approximation.
- **gamma_soft_75/50** ($\alpha = 0.75, 0.50$): soft interpolation $g' = \alpha g_{\text{null}} + (1 - \alpha)g$.

5.2 Stage 1 Results: Audit Summary

Across all 5 seeds, 41–42 of 42 candidate parameters pass Criterion A ($A > 0.5$, ≥ 2 subsets), demonstrating that Γ^{info} near-null structure is robust and reproducible throughout GPT-2 layers h.6–h.11. The overall pass rate is 98.6% (207/210 instances across 5 seeds $\times$ 42 params). Failures are concentrated in `mlp.c_fc.weight` (pass rate 90%, 3 failures across seeds 456 and 1234); all other parameter types achieve 100%. The 90% type-level rate translates to at most 1–2 failed parameters per seed (out of 6 `mlp.c_fc.weight` instances), consistent with the per-seed count of 41–42 passing parameters reported above.

Table 1: Audit pass rate by parameter type (aggregated over 5 seeds $\times$ 6 layers = 30 instances per type). A: mean cross-batch alignment score for passing instances. Pass rate computed as (passing instances) / (total instances across all seeds). Mode: *full_t2* = standard two-pass autograd; *hybrid_fd* = finite-difference correction for Conv1D parameters with near-zero HVP rows.

Parameter type	n/seed	Pass rate	A (mean)	Mode
ln.weight (d=768)	12	100%	0.998	full_t2
attn.c_attn.bias (d=2304)	6	100%	1.000	hybrid_fd
attn.c_proj.bias (d=768)	6	100%	0.926	full_t2
mlp.c_fc.weight (d=2.36M)	6	90%	0.963	full_t2
mlp.c_fc.bias (d=3072)	6	100%	0.889	full_t2
mlp.c_proj.bias (d=768)	6	100%	0.945	full_t2
Overall	42	**98.6%**	**0.953**	

The $A \approx 1.0$ for layer-norm and attention parameters confirms that Γ^{info} null-space structure is maximally reproducible across batch splits for these parameter types. The hybrid_fd mode for `attn.c_attn.bias` (Conv1D silent-zero HVP, requiring finite-difference correction) achieves identical pass rates with $A = 1.000$, validating the two-pass hybrid correction.

[1] Standard deviations are population std (divisor $N = 5$), consistent with the summary statistics reported by the experimental script.

5.3 Stage 2 Results: Continual Learning (5 Seeds)

Table 2: CL results (WikiText-2 → Biomedical Medical QA) averaged over seeds $\{42, 123, 456, 789, 1234\}$. Δppl_A: Task A forgetting (mean ± std; lower is better). ppl_B: Task B perplexity (mean; lower is better). Δ: difference in mean forgetting vs. free. Significance: paired two-tailed t-test vs. free. n_{pass}: mean number of audit-gated parameters.

Method	Δppl_A ($\mu \pm \sigma$)	ppl_B	Δ	sig	n_{pass}
free	$+414 \pm 45$	9.54	0	—	—
ewc	$+217 \pm 33$	9.48	-197	**	—
packnet_simplified	$+225 \pm 11$	9.41	-189	**	—
oEWC	$+975 \pm 136$	86.63	$+561$	**	—
gamma_along	$\mathbf{+331 \pm 30}$	**9.55**	$\mathbf{-83}$	*	**41**
gamma_random	$+369 \pm 68$	9.68	-45	ns	41
gamma_soft_75	$+459 \pm 41$	9.76	$+45$	ns	41
gamma_soft_50	$+468 \pm 39$	9.75	$+54$	ns	41

$**$ $p<0.01$; $*$ $p<0.05$; ns: not significant. Paired two-tailed t-test.

Table 3: Per-seed forgetting Δppl_A for key methods.

Method	s=42	s=123	s=456	s=789	s=1234	mean±std
free	440	374	489	396	372	414 ± 45
ewc	238	170	247	184	246	217 ± 33
packnet_simpl.	224	219	245	213	226	225 ± 11
gamma_along	353	369	329	323	282	331 ± 30
gamma_random	470	306	302	340	427	369 ± 68

Key findings. (1) **Significant forgetting reduction.** `gamma_along` reduces forgetting significantly vs. free (+331 vs. +414, $\Delta = -83$, $p<0.05$), a 20% reduction. Task B perplexity is preserved (9.55 vs. 9.54), confirming that null projection does not compromise new-task learning.

(2) **Direction quality signal.** `gamma_along` < `gamma_random` (331 vs. 369, $\Delta = -38$, ns). The gap is directionally consistent with the geometric claim—audit-identified null directions, not merely projection, contribute to forgetting reduction—but does not reach statistical significance at 5 seeds.

(3) **Comparison to stronger baselines.** EWC and PackNet achieve lower absolute forgetting ($\approx +220$) via full Fisher importance weighting and 50% weight freezing respectively. `gamma_along`'s contribution is not to match these strong regularizers, but to demonstrate that a purely geometric null-space projection—using only audit-gated parameter directions and no additional loss-landscape computation—can significantly reduce forgetting ($p<0.05$) while preserving Task B performance. Further closing the gap to EWC/PackNet is the natural next step, via curvature-weighted projection (see §7).

(4) **Hard vs. soft projection.** Soft variants ($\alpha = 0.75, 0.50$) perform worse than free, confirming that partial projection is insufficient: either full projection ($\alpha = 1.0$) or no projection is preferred at this scale.

(5) **oEWC instability.** Online EWC diverges at lr $= 5 \times 10^{-4}$ ($\text{ppl}_B = 86.6$, forgetting +975), demonstrating that it is not robust to the aggressive learning rates typical in modern fine-tuning. All methods use the same learning rate for a fair comparison; this setting is unfavorable

to oEWC but reflects realistic fine-tuning practice.[2]

6 Analysis: Direction Quality and Projection Strength

Two aspects of the results warrant deeper analysis: the gap between `gamma_along` and `gamma_random` (which bears on the geometric claim that direction quality matters), and the hard/soft projection tradeoff (which bears on practical deployment).

6.1 gamma_along vs. gamma_random Gap

The key ablation—`gamma_along` vs. `gamma_random`—yields $\Delta = -38$ (ns at 5 seeds). This gap is directionally consistent with Case A (direction quality matters) but does not reach statistical significance at the current sample size.

Notably, `gamma_random` exhibits substantially higher variance ($\sigma = 68$) than `gamma_along` ($\sigma = 30$). This difference suggests that audit-identified null directions are not only better on average but also more stable across seeds, consistent with a genuine geometric signal rather than random fluctuation. A random projection basis produces outcomes that vary widely depending on which random directions happen to align with the forgetting structure of a given seed; the audit-gated basis converges to a reproducible geometric subspace that yields more consistent forgetting reduction. Expanding to more seeds or additional Task B domains would provide the statistical power needed to confirm whether the direction gap reaches $p<0.05$.

6.2 Hard vs. Soft Projection

Soft variants ($\alpha = 0.75, 0.50$) perform consistently worse than free, while hard projection ($\alpha = 1.0$) is the only variant that reduces forgetting. This suggests that at the scale of GPT-2 with aggressive lr $= 5 \times 10^{-4}$, partial null-space projection provides insufficient constraint—the gradient component orthogonal to the null space dominates and still causes forgetting. Full projection ($\alpha = 1.0$) may be the appropriate operating point for this regime, while softer variants may become effective at smaller learning rates or larger models where individual parameter updates are smaller.

Full projection does not impede Task B learning (Table 2: $\text{ppl}_B = 9.55 \approx$ free 9.54), consistent with null directions remaining open for Task B gradient flow.

7 Curvature-Aware CL: A Roadmap

The results above establish Γ^{info} as a valid geometric proxy for audit-gated null-space projection in GPT-2. We now describe three curvature-aware CL methods that are directly derivable from the field equation and represent the natural next experimental steps. None are evaluated here; they constitute a principled research program.

7.1 Curvature-Aware Layer Selection

The field equation predicts, prior to any experiment, which layers are geometrically stable CL targets:

$$\text{select layers where } \|\Gamma_l^{\text{info}}\| < \tau_{\text{layer}}.$$

Layers with low curvature are geometrically stable (§3.2) and their null spaces are more reliable (higher A; Table 1). This replaces the current multi-layer audit with a single curvature-threshold selection step.

[2] oEWC is known to be sensitive to learning rate; a tuned lower lr would likely stabilize it, but would also reduce its Task B learning speed.

7.2 Curvature-Aware Learning Rate Scaling

The field equation directly motivates a per-layer LR schedule:

$$\eta_l = \frac{\eta_0}{1 + \alpha \|\Gamma_l^{\text{info}}\|}, \tag{4}$$

where $\alpha > 0$ is a scaling coefficient. High-curvature layers receive smaller learning rates, reducing forgetting at geometrically unstable layers; low-curvature layers retain full learning capacity. Li [12] show that geometry-guided per-layer LR achieves $\Delta = +2.7 \pm 1.1\%$ over uniform LR on ViT-Tiny, validating the $\|\Gamma^{\text{info}}\|$-based signal. Eq. 4 is the simplest instantiation compatible with standard optimizers.

7.3 Curvature-Weighted Null-Space Projection

The current projection is binary: either a parameter is projected or it is not. A natural extension weights projection by geometric instability:

$$\Delta\theta_l \leftarrow w_l \, P_{N_l} \, \Delta\theta_l + (1 - w_l) \, \Delta\theta_l, \quad w_l = \frac{1}{1 + \alpha \|\Gamma_l^{\text{info}}\|}, \tag{5}$$

where P_{N_l} is the null-space projector for layer l. When $\|\Gamma_l\|$ is large, $w_l \to 0$ and the update is barely projected (the null space is unreliable); when $\|\Gamma_l\|$ is small, $w_l \to 1$ and the update is fully confined to the null space. This combines null-space projection with field-equation guidance in a single differentiable operation.

7.4 Connection to Current Results

The current `gamma_along` method uses a binary audit gate (pass/fail based on Criterion A) with full projection ($\alpha = 1.0$). The significant forgetting reduction ($\Delta = -83$, $p<0.05$) establishes that audit-gated null projection is empirically effective at this scale. The curvature-weighted extension (Eq. 5) would further modulate projection strength by layer geometry—predicted to improve the hard/soft tradeoff observed in Table 2.

8 Related Work

Continual learning. Kirkpatrick et al. [4] introduced EWC; Schwarz et al. [6] extended it to online Fisher estimation (oEWC). Zenke et al. [11] proposed synaptic intelligence; Mallya & Lazebnik [5] introduced PackNet, which reserves weight capacity per task through iterative pruning. Lopez-Paz & Ranzato [7] and Farajtabar et al. [2] develop gradient-space projection methods. Our work is most closely related to the projection family: we project gradients onto audit-gated index-subspace null approximations of Γ^{info}, with a matched random-direction control that isolates direction quality from projection geometry. Unlike EWC-family methods, our projection requires no per-step penalty computation; unlike PackNet, it requires no iterative task-specific reallocation. We compare against representative regularization- and masking-based baselines (EWC, oEWC, simplified PackNet), but do not claim exhaustive SOTA coverage; rehearsal-based methods (GEM, A-GEM) are not included.

Null-space methods. Li [10] introduced NCCL (Neural Null Cones), which defines null directions via a symplectic condition on the gradient covariance matrix; those directions are structurally defined but lack an information-theoretic derivation. Zeng et al. [13] use orthogonal weight modification. Neither provides a pre-CL diagnostic for null-space quality. Our audit framework provides such a diagnostic via a cross-batch reproducibility criterion (Criterion A): only parameters whose Γ^{info} near-null eigenspaces are stable across disjoint batch halves enter gradient projection, grounded in the entropy curvature tensor [12].

Information geometry. Amari [1] and Martens [8] apply Fisher–Rao geometry to optimization. Our Γ^{info} targets the entropy Hessian rather than the Fisher metric, and focuses on null rather than curvature directions.

Information field equation. Li [12] derive and experimentally validate the field equation $G \approx -2\, dt^2_{\text{info}}\, \Gamma^{\text{info}}$ on ViT-Tiny and GPT-2, showing that the entropy curvature term T_2 dominates $\|G\|$ by 87–114% in the entropy-dominated regime. They further demonstrate that $\|\Gamma^{\text{info}}\|$ predicts per-layer forgetting (Spearman $\rho = 0.822 \pm 0.054$) and that geometry-guided per-layer learning rates improve over uniform schedules ($\Delta = +2.7 \pm 1.1\%$). Our work extracts the *near-null subspace* of Γ^{info} as safe CL update directions via index-subspace audits across GPT-2 layers h.6–h.11; the field equation's degradation at boundary blocks [12] motivates treating lower A scores at outer layers as expected rather than as failures.

9 Conclusion

We proposed an *audit-gated gradient projection family* for continual learning, grounded in the information field tensor Γ^{info} [12]. Rather than claiming exact null-space recovery, we introduced a Gamma cross-batch reproducibility audit (Criterion A, Eq. 3) that verifies null-space stability across disjoint batch halves before projection.

In a cross-domain experiment (GPT-2, WikiText-2 → Biomedical Medical QA, 5 seeds): (i) the audit gates 41–42 of 42 candidate parameters across all seeds ($A \geq 0.71$), confirming that Γ^{info} near-null structure is real, reproducible, and present throughout GPT-2 layers h.6–h.11; (ii) audit-gated null projection (`gamma_along`) significantly reduces forgetting ($+331 \pm 30$ vs. free $+414 \pm 45$, $\Delta = -83$, $p<0.05$) without sacrificing Task B performance (Table 2); (iii) `gamma_along` outperforms matched random projection (§6), supporting the geometric claim that audit-identified null directions—not merely projection—matter.

Claim layering. The current result is Case A: `gamma_along` < `gamma_random` < free, supporting that null direction quality contributes to forgetting reduction, though the `gamma_along` vs. `gamma_random` gap is not yet significant at 5 seeds. `gamma_along` does not match EWC/PackNet in absolute forgetting reduction; the contribution is geometric and principled rather than empirical SOTA.

Limitations.

1. **Index-subspace approximation only.** We do not recover the exact full-parameter null space of Γ^{info}. Criterion A verifies cross-batch reproducibility of k-dimensional sub-block near-null structure, not a global geometric certificate.

2. **Simplified baselines.** `packnet_simplified` is a one-shot magnitude mask. Rehearsal-based methods (GEM, A-GEM) and adapter-based approaches are not included.

3. **Scale and domain.** Experiments use GPT-2 (124M) on a single cross-domain pair. Scalability to larger models and multiple task sequences is future work.

4. **Direction gap not yet significant.** `gamma_along` vs. `gamma_random` ($\Delta = -38$, ns) supports but does not statistically confirm the direction quality claim at 5 seeds.

5. **oEWC instability.** oEWC diverges at lr $= 5 \times 10^{-4}$; incompatibility with aggressive learning rates is a known limitation.

Broader impact. A geometry-grounded stability signal could enable developers to predict, prior to deployment, which directions in parameter space are safe for continual task adaptation—a form of *geometric alignment checking* before fine-tuning begins.

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